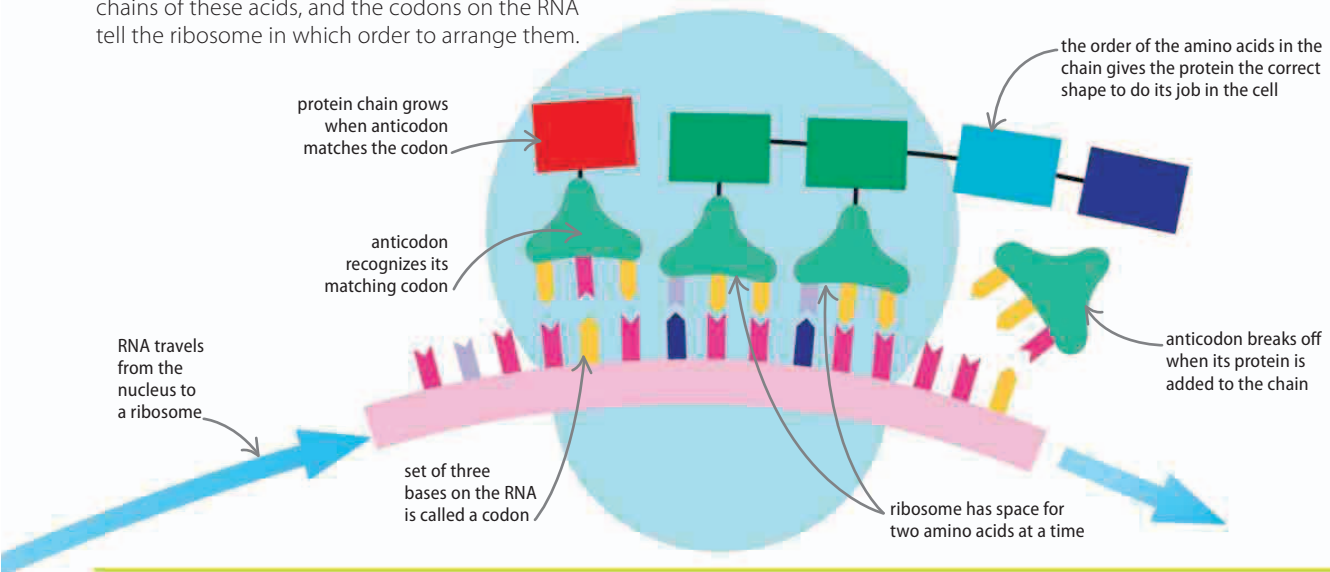


Translation

The genetic code is translated into a protein in the cell's ribosome. This tiny organelle pulls the RNA through itself three bases at a time. Every three bases form a triple-character sequence called a codon, which is specific to a certain amino acid. Proteins are composed of chains of these acids, and the codons on the RNA tell the ribosome in which order to arrange them.

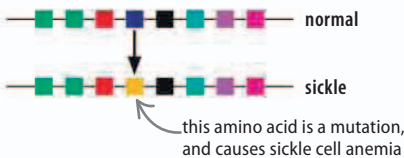
▽ Making proteins

Anticodons carry specific amino acids to the ribosome to add to the chain. When the anticodon matches the codon on the RNA, the anticodon adds its amino acid to the chain, and the next codon is pulled into the ribosome.



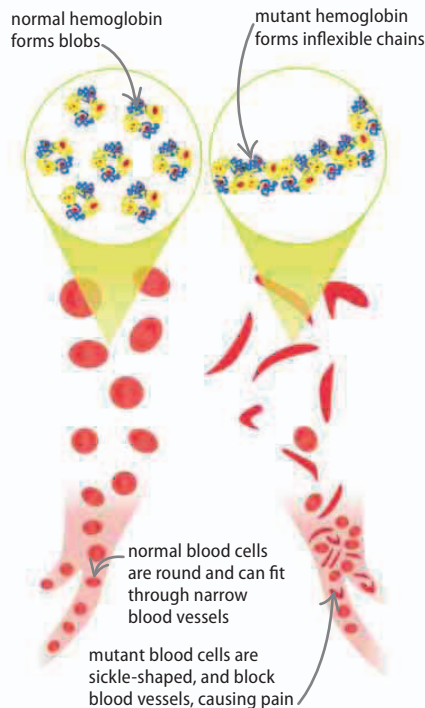
Mutations

When DNA is copied, mistakes can occur. These are called mutations. A mutation may be made in the unread part of DNA, and so have no effect. If it happens in the read section, the result can make the cell die. However, occasionally a mutation improves the way the cell and the body works. These useful mistakes are spread by natural selection and drive evolution.



△ Genetic disease

Some mutations are not deadly, but cause diseases. For example, sickle cell anemia is caused by one different amino acid in the structure of hemoglobin, the chemical that carries oxygen in the blood. The mutant hemoglobin forms long chains, which makes a sufferer's blood cells sickle-shaped.



REAL WORLD

Human genome project

A genome is the complete collection of a species' genes. In 2003, scientists finished a complete record of the human genome. They identified about 25,000 genes and sequenced three billion base pairs. Below is a section of the genome, with a color for each base. However, geneticists have still to figure out what most of the genes do and record their many different versions, or alleles.

